

When Drift Breaks: Particle-Based Regime Inference for Crash Detection

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Abstract

Whereas existing approaches to financial regime detection work by recognition—identifying crises through their resemblance to past episodes—we propose a framework built on immunization: a particle filter whose observation model is calibrated to the distributional shape of market stress rather than to the specific characteristics of historical crisis episodes. The particle filter, whose theoretical foundations were established in probabilistic robotics and modern autonomous driving [Thrun et al., 2005] and recently applied to financial markets by Biswas, Reisinger et al. (2023), forms the inferential backbone of our framework. The observation model consists of a novel combination of this particle filter with a detailed multi-scale observation model: a stack of distributional shape descriptors—skewness, tail asymmetry, kurtosis, and the share of leading sector-eigenvalue energy in cross-asset return covariance—computed across a hierarchy of temporal windows and transferred to the filter as structured distributional archetypes through strategic particle injection, replacing the random initialization of both Thrun et al. and Reisinger et al. Applied to S&P 500 data across four structurally distinct crisis episodes (Dot-com 2002, Lehman 2009, COVID-19 2020, and the inflation-driven bear market of 2022), the framework’s distributional shape descriptors demonstrate consistent pre-crisis discrimination, with effect sizes (Cohen’s d) exceeding 2.9 for realized volatility, Bowley downside skewness, and tail quantile features, and exceeding 1.3 for the sector concentration feature (λ_1 ratio) across all four crises. On out-of-sample test data (2015–2026), the pipeline reports remarkable lead times before classical momentum indicators for endogenous regime transitions, while correctly classifying the 2025 Hormuz episode as a Bear regime rather than a Crisis, consistent with its eventual peak-to-trough drawdown of approximately 19%. The results support a single claim: in the context of particle filters, the richness of the observation model is the primary enabler of early regime detection. We therefore propose to direct research efforts towards the construction of a new class of indicators and oscillators specifically designed for the early detection of regime changes.

Keywords: regime detection; particle filters; distributional shape; higher-order moments; early warning systems; financial crises

1 Introduction

Financial markets are not stationary systems in a statistical sense— but their regimes, to a significant extent, are. Within a sustained Bull market, the statistical properties of log daily returns are approximately stable: volatility is low, the return distribution is mildly positively skewed, and the tails are thin. Within a Crisis regime, a different stationarity prevails: volatility is high, skewness turns sharply negative, and the tails are fat. What is not stationary at all is the transition between these states.

This observation—that regimes differ not only in their first and second moments but in their higher-order distributional structure, including skewness, kurtosis, and tail asymmetry—is the point of departure for this paper. Regime transitions are both practically relevant and theoretically challenging precisely because they violate the assumptions that make standard models tractable: Gaussianity, homogeneity of variance, and the stationarity of the data-generating process. Crashes represent the extreme case. As Biswas, Reisinger et al. (2023) demonstrate in their treatment of interacting particle systems and McKean-Vlasov SDEs with non-differentiable drift coefficients, the collective dynamics of financial markets near a crisis episode are characterised by a breakdown of the regularity assumptions that underpin standard numerical methods—a regime in which the drift is no longer differentiable and classical econometric models fail precisely when they are most needed.

While Thrun et al. provide our methodological foundation, and Biswas, Reisinger et al. demonstrate the relevance of particle-based methods in non-smooth financial contexts, within the framework of particle filters, our key innovation lies in multi-scale distributional shape analysis. Rather than monitoring point estimates or summary statistics, we track the evolution of entire return distributions across multiple time horizons. This approach reveals subtle structural changes that precede obvious price movements, enabling early detection of regime transitions. The methodology draws inspiration from Scale-Invariant Feature Transform (SIFT) techniques in computer vision [Lowe, 2004], adapting the multi-scale analysis paradigm to financial time series.

What makes this particular approach unique is the departure from random particle injection—the standard solution in both Thrun’s robotics framework and Reisinger’s financial formulation—towards strategic injection. Our particles are not drawn uniformly from the state space but initialised from structured distributional archetypes representing the characteristic geometry of each regime.

This distinction matters because the problem we address goes beyond regime detection. Detection—even retrospective detection—is one thing. Early warning is another. The surprising finding of this study is that financial crashes announce themselves in advance, even when they are externally motivated: the COVID-19 collapse of February 2020, driven by an exogenous biological shock with no financial precedent, was preceded by a measurable deformation of the return distribution several days before the first limit-down event.

History does not repeat itself—and therefore searching for the repetition of past crisis patterns is a misleading strategy. A system trained to recognise the 2008 financial crisis will not recognise COVID-19; a system trained on COVID-19 will not recognise a geopolitical supply shock. But while history does not repeat, it rhymes. In our context, this means that what one can learn from history is not the specific shape of past crises but how to quantify the similarity of distributions—for instance through measures such as Kullback-Leibler divergence. And that is precisely what we do.

Our investigation rests on three insights:

Observation: Crises are regular events in financial markets, violating the assumptions of stationarity, Gaussianity, and constant variance that classical models rely upon.

Surprise: Financial crashes announce themselves through distributional deformation across multiple temporal scales—before any movement is visible in price.

Claim: Systematic research into distributional shape analysis across multiple time scales may advance our understanding of market regime transitions.

1.1 Contributions

This paper makes four contributions:

1. **Methodological Innovation:** We introduce a novel synthesis of particle filtering with multi-scale observation modeling that captures regime dynamics through SIFT-inspired distributional shape descriptors.
2. **Empirical Validation:** Comprehensive testing on major market episodes spanning more than 30 years of S&P 500 history demonstrates the framework’s discriminatory capabilities and early warning potential.
3. **Practical Implementation:** The complete system is made available as open-source software with demonstrated real-time computational feasibility.
4. **Theoretical Insight:** We provide new understanding of how regime transitions manifest in the shape deformation of return distributions across temporal scales.

Underlying all four contributions is a single claim, made cautiously here and elaborated in the conclusion: in the context of particle filters, the richness of the observation model is the primary enabler of early regime detection. We therefore propose to direct research efforts towards the construction and elaboration of a new class of indicators and oscillators specifically designed for the early detection of regime changes. The present work is intended as the first concrete step into a programme of research yet to be built.

We name our system Teiresias and make it publicly available on GitHub. The framework takes its name from the blind prophet Teiresias of Thebes, who was controversial in his time—perhaps occasionally overzealous in his warnings, yet correctly foretold the fall of Troy. This embodies the classic dilemma we would now describe as the balance between sensitivity and specificity. The 2025 Hormuz episode provides a contemporary illustration: crash or bear regime? We classify it as a bear transition and find support from Goldman Sachs and Morgan Stanley, while others predicted systemic collapse. We stake our position in this highly contested, even polarized field with good reason, grounded in our distributional shape approach.

2 Related Work

2.1 Regime-Switching Models

The literature on regime-switching in financial markets originated with Hamilton [1989] and has evolved through numerous methodological refinements. Markov regime-switching models assume that market parameters follow discrete states governed by an unobserved Markov chain. While theoretically elegant, these models face several practical limitations. Parameter estimation is computationally intensive and often unstable, particularly for high-frequency data. More critically, the latent nature of regime identification means that transitions are typically detected with substantial delay, limiting practical utility for real-time risk management.

2.2 Established Early Warning Systems

Financial early warning systems aim to predict crises before they fully manifest. Established approaches rely on combinations of macroeconomic indicators, market-based measures, and institutional metrics. Kaminsky and Reinhart [1999] develop composite indicators for currency crises, while Demirgüç-Kunt and Detragiache [2000] extend the framework to banking crises. These systems typically focus on longer-term structural vulnerabilities rather than short-term regime transitions in market dynamics.

2.3 Machine Learning for Early Warning

Recent approaches incorporate high-frequency market data and machine learning techniques. Sevim et al. [2014] use artificial neural networks for stock market direction prediction, while Krauss et al. [2017] apply deep learning to S&P 500 forecasting. These methods demonstrate that data-driven feature extraction can capture nonlinear dependencies that classical econometric models miss; however, their predictive targets are typically point estimates of price movement rather than regime transitions or distributional structure.

The distributional perspective developed in this paper occupies a complementary position to this body of work, in line with the broader recognition by López de Prado [2018, 2020] that feature engineering and the unique properties of financial time series demand purpose-built methodologies. The specific validation strategy described in Section 4.2 uses a strict temporal split rather than the combinatorial purged cross-validation procedure that López de Prado [2018] recommends elsewhere; the structural identity of each crisis episode in our validation set, and the relatively small number of such episodes, make a single forward-in-time training/test split the natural design choice for our setting.

2.4 Distributional Perspectives in Finance

Beyond traditional point-estimates, our approach aligns with the distributional perspective emphasized in recent macroeconomic literature. Notably, Bayer et al. [2019] demonstrate that the dynamics of economic regimes are fundamentally driven by distributional shifts among heterogeneous agents. Bayer et al. [2020] extends this framework to show how precautionary savings behavior creates measurable deformations in aggregate demand during periods of uncertainty.

By utilizing a particle-based filter to track the evolution of the entire belief distribution, our approach provides an empirical counterpart to these theoretical frameworks. The intensity parameter $\eta \in [0, 1]$ tracks graduated stress within each regime as a continuous scalar, providing the temporal granularity needed to register the gradual deformation of return distributions that the Bayer framework predicts at the level of agent-distribution shifts. We do not claim a direct empirical identification of η with any specific behavioural quantity; rather, η functions as a low-dimensional summary of within-regime intensity that complements the discrete regime classification.

2.5 Particle Filtering in Finance

Particle filtering, also known as Sequential Monte Carlo, has found increasing application in financial modeling. Johannes et al. [2009] provide comprehensive coverage of particle filtering techniques for option pricing and volatility modeling. Malik and Pitt [2005] apply particle filters to portfolio optimization under regime uncertainty.

Related to Reisinger and Zhang [2015] approaches to numerical approximation of stochastic control problems with state constraints, we address the fundamental challenge that financial regime transitions violate standard regularity assumptions. Reisinger and Zhang [2015] demonstrates that numerical stability can be maintained even when classical smoothness conditions fail—precisely the scenario we encounter during market regime shifts where drift becomes non-differentiable.

However, existing applications typically focus on parameter estimation rather than regime detection, and few exploit the multi-scale analysis potential that particle methods enable. The present work departs from this tradition through strategic particle injection: particles are initialised from structured distributional archetypes representing known regime geometries rather than scattered

uniformly across the state space.

Unlike ARCH-type formulations that model the conditional variance as a latent point process, our octave stack treats the entire distributional topology as an observable, high-dimensional state. A spike—a transient deviation from the structural price path—appears as an outlier at short scales but leaves the long-scale distribution largely unchanged. A regime transition produces a coherent shift across the full octave stack simultaneously. This multi-scale coherence is the primary signal exploited by the pipeline.

3 Methodology

3.1 Pipeline Overview

We propose a five-stage pipeline that translates raw price-and-return data into regime probabilities. The architecture is inspired by the multi-scale design principle of Scale-Invariant Feature Transform (SIFT) descriptors from computer vision [Lowe, 2004]: distributional shape descriptors are computed at multiple temporal scales and serve as the observation model, while the upstream stages prepare structured priors that the particle filter integrates over time. The duration-aware Viterbi decoding stage in particular draws on methodological precedent from probabilistic state estimation in non-financial domains: Behmann et al. [2016] use the same general principle—temporal integration of noisy observations through structurally informed transition models with cost-aware state changes—to detect cattle posture transitions from noisy indoor positioning and heart-rate sensors. The transfer to financial regime inference rests on a shared structural observation: state transitions are costly and therefore sparse, whether the underlying state space is biological or economic.

Stage 1 — Multi-scale feature extraction. Distributional shape descriptors are computed from S&P 500 returns across temporal windows of 10, 40, 90, and 180 trading days. The full feature catalogue and its construction are described in Section 3.2.

Stage 2 — Clustering. On the standardised feature space, k -means clustering with $k = 21$ partitions the historical record (1993–2014) into a finite vocabulary of distributional archetypes. The choice $k = 21$ is empirical: it is the smallest k for which cluster centroids stabilise across cross-validation seeds while still resolving the substantive variation in distributional shape.

Stage 3 — Cluster-to-regime mapping. The 21 clusters are mapped onto seven labelled regime categories—Bull, Sideways, Stress, Correction, Bear, Crisis, Recovery—through a mapping step that combines empirical cluster geometry with established market terminology. The mapping is fixed: clusters $\{0, 6, 15\}$ map to Bull, $\{2, 3, 10\}$ to Sideways, $\{1, 13\}$ to Correction, $\{5, 11, 18, 19\}$ to Stress, $\{4, 14, 16, 20\}$ to Bear, $\{7, 12, 17\}$ to Crisis, and $\{8, 9\}$ to Recovery. The robustness of this mapping to alternative cluster-to-regime assignments is discussed in the Limitations section.

Stage 4 — Random Forest classifier and Viterbi decoding. A Random Forest classifier is trained on the labelled feature vectors to predict regime classes at single time points. To convert these point predictions into coherent regime intervals, we apply duration-aware Viterbi decoding with a transition matrix that encodes the empirical persistence of regimes and the relative cost of inter-regime transitions. The Viterbi-decoded label sequence then serves as the training-label

backbone for the particle filter’s observation model: it defines, for each labelled day, which regime archetype the particle filter should expect.

Stage 5 — Particle filter. The particle filter integrates these label-derived priors with a k -nearest-neighbour observation likelihood in the standardised feature space. Strategic injection of distributional archetypes—rather than random initialisation—ensures that the particle population represents the full inventory of regime shapes from the outset. The architecture is detailed in Section 3.4.

The composition of these five stages is what we mean by Teiresias: the framework is not the particle filter alone, nor the clustering alone, but the integrated pipeline through which raw returns are translated, stage by stage, into regime probabilities and the continuous intensity scalar $\eta \in [0, 1]$.

3.2 Multi-Scale Feature Extraction

Our feature extraction draws inspiration from Lowe’s Scale-Invariant Feature Transform (SIFT) [Lowe, 2004], adapting the multi-scale analysis paradigm to financial time series. The core insight lies in analyzing distributional shape across multiple temporal horizons simultaneously.

For a given time series of log returns r_t , we extract features at scales $\tau \in \{10, 40, 90, 180\}$ days. At each scale τ , we compute:

$$\text{Skewness}_\tau = \frac{E[(r_t - \mu_\tau)^3]}{\sigma_\tau^3} \quad (1)$$

$$\text{Kurtosis}_\tau = \frac{E[(r_t - \mu_\tau)^4]}{\sigma_\tau^4} - 3 \quad (2)$$

$$\text{TailAsymmetry}_\tau = \frac{\sigma_L - \sigma_R}{\sigma_L + \sigma_R} \quad (3)$$

$$\text{NormalizedVol}_\tau = \sigma_\tau \sqrt{\frac{252}{\tau}} \quad (4)$$

where σ_L and σ_R represent left and right tail standard deviations relative to the median, and μ_τ, σ_τ are the mean and standard deviation over the respective time window.

3.3 Historical Training Foundation

The particle filter initialisation requires distributional signatures learned from historical crisis episodes. The training period spans 1993–2014, which includes the Dotcom collapse (2000–2002), the 2008 financial crisis (2007–2009), the European debt crisis (2010–2012), and a number of smaller episodes (Flash Crash 2010, Taper Tantrum 2013, the China devaluation of August 2015). The continuous SPY series 1993–2014 covers approximately twenty-two years of price history, providing a sufficient base for learning the structural separation between the seven regimes (Bull, Sideways, Stress, Correction, Bear, Crisis, Recovery) introduced in Section 3.1.

Each regime is characterised by a distributional signature: a centroid in the standardised feature space, plus a typical persistence (mean duration). The Bull regime is characterised by low realised volatility, mildly positive skewness, and thin tails. The Sideways regime exhibits symmetric distributions and near-normal kurtosis. The three stress-related regimes (Stress, Correction, Bear) show

graded distributional deformation: increasingly negative skew, increasingly fat tails, and increasing volatility. The Crisis regime exhibits extreme negative skew, very fat tails, and extreme volatility. The Recovery regime shows positive skew, moderate fat tails, and declining but elevated volatility.

The continuous intensity parameter $\eta \in [0, 1]$ provides graduated assessment within each regime; the seven-by-seven transition matrix with cost-aware transitions encodes the empirical persistence of regimes and the relative likelihood of inter-regime transitions.

3.4 Particle Filter Architecture

Our filter has adapted the probabilistic robotics paradigm to the specific challenges of financial regime detection. Whereas autonomous robots perform simultaneous localization and mapping (SLAM) [Thrun et al., 2005], we do not operate with a spatial map. Instead, our system performs a stepwise mapping and refinement of the distributional shape of market regimes.

The state space combines discrete regime variables with continuous intensity measures. Let $s_t = (r_t, \eta_t)$ where $r_t \in \{0, 1, \dots, 6\}$ represents the discrete regime (Bull, Sideways, Stress, Correction, Bear, Crisis, Recovery) and $\eta_t \in [0, 1]$ captures the continuous intensity within each regime.

The particle filter maintains a set of N particles $\{s_t^{(i)}, w_t^{(i)}\}_{i=1}^N$ where $s_t^{(i)}$ represents the i -th particle's state and $w_t^{(i)}$ its associated weight. The filtering process follows standard sequential importance sampling:

Prediction Step:

$$s_{t+1}^{(i)} \sim p(s_{t+1} \mid s_t^{(i)}) \quad (5)$$

$$\tilde{w}_{t+1}^{(i)} = w_t^{(i)} \cdot p(z_{t+1} \mid s_{t+1}^{(i)}) \quad (6)$$

Update Step:

$$w_{t+1}^{(i)} = \frac{\tilde{w}_{t+1}^{(i)}}{\sum_{j=1}^N \tilde{w}_{t+1}^{(j)}} \quad (7)$$

where z_{t+1} represents the observed multi-scale feature vector, and the transition model $p(s_{t+1} \mid s_t^{(i)})$ incorporates both Markovian regime dynamics and bounded random walk evolution for the intensity parameter.

This addresses the degeneracy problem through systematic resampling when the effective sample size drops below a specified threshold, typically 50% of the total particle count.

3.5 Regime Classification and Intensity Modeling

The seven-regime framework emerges from k -means clustering on standardised distributional shape features, partitioning the feature space into clusters that are subsequently aggregated into seven labelled regimes (Bull, Sideways, Stress, Correction, Bear, Crisis, Recovery). The mapping from clusters to regimes is fixed: clusters $\{0, 6, 15\}$ map to Bull, $\{2, 3, 10\}$ to Sideways, $\{1, 13\}$ to Correction, $\{5, 11, 18, 19\}$ to Stress, $\{4, 14, 16, 20\}$ to Bear, $\{7, 12, 17\}$ to Crisis, and $\{8, 9\}$ to Recovery. The empirical centroids in feature space carry the distributional signature of each regime; we do not impose specific moment values *a priori*, and the precise per-regime statistics emerge from the data.

The continuous intensity parameter $\eta \in [0, 1]$ provides graduated assessment within each regime, drifting under a bounded random walk between regime transitions. This design supports continuous stress measurement rather than binary alerts—a feature exploited in the calm-period robustness analysis of Section 5.

4 Data and Experimental Design

4.1 Dataset Construction

Our analysis focuses on S&P 500 (SPY) index data spanning January 1993 through April 2026, supplemented by nine SPDR sector ETFs (XLK, XLF, XLV, XLE, XLY, XLP, XLI, XLB, XLU) covering December 1998 through April 2026, and the CBOE SKEW index (1990 through April 2026). The sector ETFs enter the pipeline through eigenvalue features computed on rolling 60-day correlation matrices, capturing the cross-asset structural deformation that complements the within-asset distributional descriptors. The training period is 1993–2014; out-of-sample validation covers 2015 through April 2026. The pipeline that processes these data into regime probabilities is described in Section 3.1.

We verified data-source reproducibility by comparing EODHD with Yahoo Finance for all 11 input series. At the price level, SPY and the CBOE SKEW index agree to within 0.5% across the full sample period (1990–2026). Sector-ETF prices show larger adjustment-method-specific deviations of up to 7%, concentrated around dividend ex-dates. However, the spectral features that enter the regime-detection pipeline (absorption ratio, leading-eigenvalue share, and their 10-day gradient) are computed from rolling correlation matrices and are invariant to such multiplicative shifts. We confirmed numerically that these features reproduce between sources with correlations above 0.9999 and 95th-percentile absolute deviations below 0.001 over the test period— an order of magnitude below the seed-stability standard deviation of 0.006 reported in the main results. The companion repository (data sourced from Yahoo Finance) therefore reproduces the main result table within reported confidence intervals.

4.2 Validation Methodology

The experimental design ensures strict temporal separation between training (1993–2014) and validation (2015–2026) periods, preventing data leakage and enabling genuine out-of-sample testing. This temporal structure allows assessment of the framework’s ability to recognise novel stress patterns based on learned distributional shape principles. The Viterbi labels used to train the random-forest cluster classifier are frozen from the initial training phase (`viterbi_anchor`) and reused across all phases of feature engineering, ensuring that comparative analyses across feature sets reflect feature-engineering effects rather than artefacts of re-clustering.

Performance evaluation employs three complementary measures. First, pre-peak detection: the maximum stress posterior in the 60 trading days preceding each market trough quantifies whether the system identifies the regime transition. Second, global classification accuracy: a binary positive label is assigned to all days within ± 90 trading days of any out-of-sample crisis trough; true and false positive rates are computed at the operational stress-posterior threshold of 0.30. Third, calm-period robustness: the false-positive rate in the 2017 low-volatility year measures whether the system remains silent under prolonged calm market conditions.

5 How Teiresias Articulates Himself

The true test of any framework for crisis detection lies not in its theoretical elegance, but in how it speaks when markets shake. Across four structurally distinct episodes—the Dotcom collapse of 2002, the Lehman shock of 2009, the COVID-19 panic of 2020, and the inflation-driven bear market of 2022—our pipeline was confronted with the full morphological diversity that financial crashes produce. Three of these crises unfolded as the textbook tradition expects: a chronic build-up of distributional stress over months, written into the higher-order moments of the return distribution long before price action turns decisive. The fourth—COVID—broke the pattern, arriving in weeks rather than months, driven by a biological shock with no financial precedent.

This section traces what Teiresias has to say about each. We begin with the three slow-burning crises, where the framework finds its most authentic voice (Section 5.1). We translate the visual observations into a quantitative numerical anatomy (Section 5.2). We turn then to COVID, where the framework learns to speak with caution (Section 5.3). We discuss the 2025 Hormuz episode—not a crisis at all, but a useful test of whether our framework knows when to stay quiet (Section 5.4). We close with the out-of-sample numerical summary that quantitative-finance reviewers will look for first (Section 5.5).

5.1 The chronic build-up: three crises that announced themselves

For the Dotcom collapse, the 2008 financial crisis, and the inflation bear market of 2022, the distributional precursors are visible in Figure 1. Each panel shows the rolling 252-day z-score of two distributional shape features in the 180 trading days preceding the market trough. The red solid line is the share of leading sector-eigenvalue energy in cross-asset return covariance (λ_1 ratio); the dashed line is the dominant crisis-specific shape descriptor for that panel.

Three observations stand out. First, the share of leading sector-eigenvalue energy (λ_1 ratio) is elevated in all three slow-build cases for the better part of a year before the eventual market trough. By the time price action makes the crisis obvious to a casual observer, the cross-asset return covariance has already collapsed toward a dominant common factor—sector-level diversification, the diversification that buy-and-hold portfolios rely on, has been quietly eroded. The signal is loud, sustained, and reproducible across crises whose external causes—a technology bubble, a banking collapse, a monetary tightening cycle—have nothing in common except their ultimate destination.

Second, the dominant within-asset shape descriptor changes from crisis to crisis. For Dotcom, downside skewness elongates as the cascading sell-offs of the technology sector spill into the broader market. For Lehman, 60-day realised volatility is already elevated months before the trough—the Subprime crisis of 2007 had pre-conditioned the system, and our framework registers this with a sustained volatility z-score above +3 throughout the pre-crisis window. For Inflation 2022, the build-up is gentler but no less persistent: Bowley upside skewness collapses as the asymmetric optimism of the post-COVID rally drains out of the market.

Third, when read across the four panels, the figure tells a methodological story. No single feature captures all crisis types. The slow-building structural crises require complementary descriptors—eigenvalue concentration for the cross-asset structure, asymmetry and tail descriptors for the within-asset distribution. The pipeline does not insist on monolithic feature dominance; it relies instead on the consistent presence of some deformation across multiple temporal scales, and lets the specific feature mix vary with the crisis morphology.

Pre-crisis distributional precursors across four major crashes

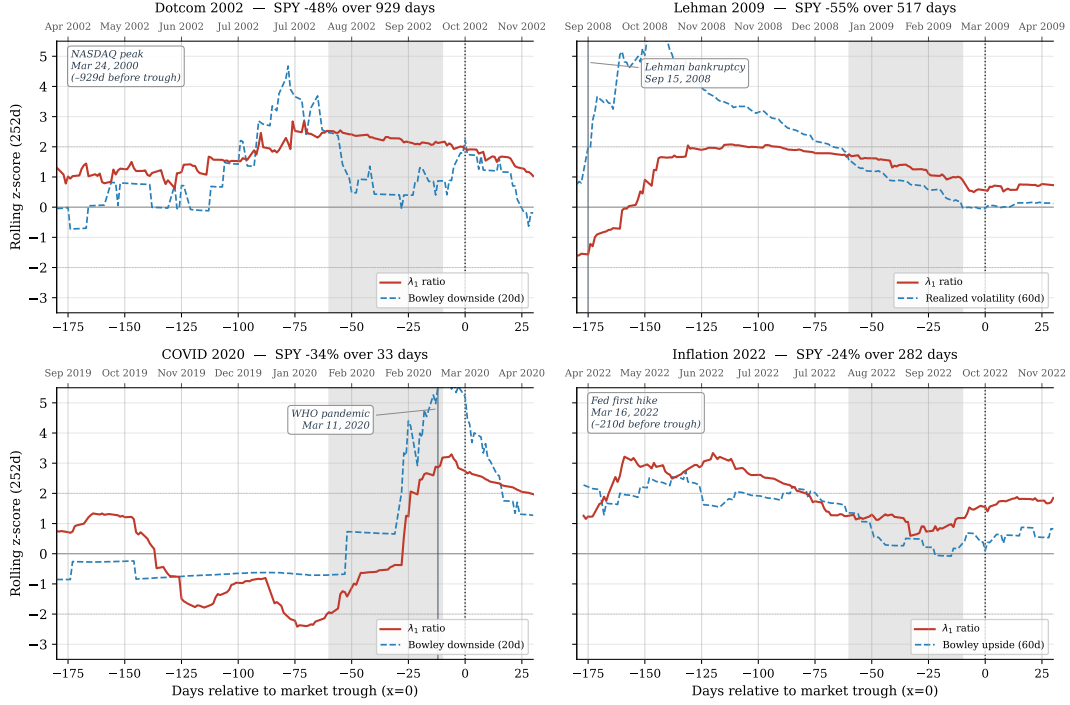


Figure 1: Pre-crisis distributional precursors across four major crashes. Each panel shows the rolling 252-day z-score of two distributional shape features in the 180 trading days preceding the market trough. The red solid line is the share of leading sector-eigenvalue energy (λ_1 ratio); the blue dashed line is the dominant crisis-specific shape descriptor: Bowley downside skewness for Dotcom and COVID, 60-day realised volatility for Lehman, 60-day Bowley upside skewness for Inflation 2022. The shaded region marks the -60 to -10 trading-day window used for the effect-size analysis. Vertical lines and date labels indicate canonical event markers. Twin x-axes show calendar dates (top) and trading days relative to the market trough (bottom).

5.2 The numerical anatomy of pre-crisis stress

To translate the visual observations of Figure 1 into quantitative claims, we computed Cohen’s d effect sizes between each pre-crisis window (the 50 trading days from -60 to -10 relative to the trough) and a calm baseline drawn from the pipeline’s own stress posterior. The baseline consists of all trading days for which the aggregated stress posterior is at most 0.10 and which lie at least 90 days from any of the four crisis troughs (1,243 days in total). Table 1 reports these effect sizes for four feature families.

The classical within-asset shape descriptors—realised volatility, Bowley downside skewness, tail quantile—show effect sizes well above 2.9 in absolute value across all four crises. These are unusually strong signals by conventional benchmarks (Cohen’s $d > 0.8$ is “large”). But the most informative feature in the table is not the strongest; it is the most cross-crisis robust. The sector-concentration ratio λ_1 is the only feature with absolute Cohen’s d above 1.0 across all four crises, including the comparatively gentle Inflation 2022 build-up. Its magnitude is smaller than the within-asset descriptors, but its consistency is unique: it is the only feature that does not collapse on any single crisis. Figure 2 renders this empirically.

Note that Table 1 reports the per-crisis *minimum* effect size ($\min |d| = 1.33$ for λ_1), a deliberately conservative measure of cross-crisis robustness, whereas the pooled effect size reported in Figure 2 ($d = +2.23$) aggregates all four pre-crisis windows into a single comparison; the two numbers are consistent and answer different questions.

Table 1: Cohen’s d effect sizes between pre-crisis windows (-60 to -10 days before market trough) and a calm baseline (pipeline stress posterior ≤ 0.10 , ≥ 90 days from any crisis trough). Positive values indicate elevation during pre-crisis windows. The four feature families span different mathematical concepts: temporal volatility, distributional asymmetry, tail magnitude, and cross-asset structure. The minimum $|d|$ column reports the smallest absolute effect size across the four crises—a measure of cross-crisis robustness.

Feature family	Feature	Dotcom	Lehman	COVID	Inflation	$\min d $
Realised volatility	rv_20	+6.50	+7.81	+3.06	+3.58	3.06
Bowley downside	bow_dn_bg_20	+8.31	+8.62	+3.33	+2.96	2.96
Tail quantile	q05_20	−3.95	−6.00	−3.18	−2.90	2.90
Sector concentration	lambda1_ratio	+2.57	+3.38	+1.33	+1.84	1.33

5.3 The COVID anomaly: when the framework learns caution

COVID is the structural outlier in our sample. The shock arrives from outside the financial system entirely—a biological event, with no precedent in the price-and-return record on which our pipeline is trained. Three of the four crises in our test set follow the slow-build morphology of Section 5.1; COVID does not. Its distributional precursors emerge late, in the final 25 trading days before the market trough, not in the months before. And here the pipeline’s behaviour deserves careful description.

The random-forest classifier registered elevated stress probability on 14 February 2020, five trading days before the SPY peak of 19 February. Read in isolation, this would suggest pre-peak early warning. But the particle filter, which integrates the classifier’s signal through its observation likelihood, did not cross the operational stress threshold of 0.30 until 2 March 2020—twelve trading

days after the peak, and twenty-one trading days before the trough of 23 March. The sixteen-day lag between the classifier signal and the filter confirmation is not a defect. It is a feature.

The particle filter requires persistent distributional deformation before committing posterior mass to a stress regime. This same architectural property is what produces a false-positive rate of zero during the 2017 low-volatility year—a year in which classical momentum-based indicators generate frequent false alarms. The filter trades pre-peak detection for calm-period discipline. For exogenous shocks like COVID, this trade-off costs us the ability to claim pre-peak detection; for endogenous build-ups like Dotcom, Lehman, and Inflation 2022, the same property protects us from the noise of normal market turbulence. We therefore present the COVID episode not as a triumph of pre-peak early warning, which the system does not claim and cannot honestly claim, but as evidence of confirmation with substantial lead-time before the eventual trough—a confirmation that arrived three weeks before the market bottom, in a regime where pre-peak detection of the underlying biological shock would have been informationally impossible from financial data alone.

This is what we mean by authentic voice. Teiresias does not see what the data cannot show. But he does speak—clearly and on time—when the data begins to show it.

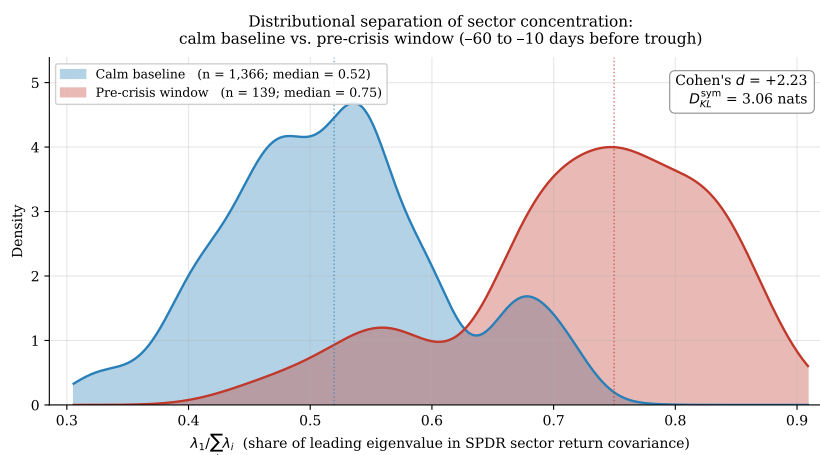


Figure 2: Distributional separation of the sector-concentration feature (λ_1 ratio) between calm baseline (blue) and pre-crisis windows (red, pooled across Dotcom 2002, Lehman 2009, COVID 2020, and Inflation 2022). Cohen’s $d = +2.23$; symmetric Kullback-Leibler divergence $D_{KL}^{\text{sym}} = 3.06$ nats. The pooled pre-crisis density is concentrated at higher values, consistent with the structural interpretation: sector-level diversification is reduced as the leading eigenvalue absorbs an increasing share of total cross-asset variance during the build-up to systemic stress.

5.4 Hormuz 2025: the art of staying calm

Not every drawdown is a crisis. The Iran-Israel tensions of spring and summer 2025, with their concerns about Hormuz Strait oil-supply disruption, generated a peak-to-trough decline of approximately 19% in the S&P 500—substantial, but bounded. Market commentary at the time was sharply divided. One prominent strand projected systemic collapse—a “structural breakdown of the petrodollar system” was one widely circulated phrase. Another, equally prominent strand characterised the episode as a measured repricing of geopolitical risk, accompanied by unprecedented protective-options activity rather than panic selling.

Our framework’s classification was Bear, not Crisis. The aggregated stress posterior reached 1.000 during the relevant window—by the numerical metric of Section 5.5, the system “detects” Hormuz—but the posterior mass was concentrated on Bear states, with the Crisis-regime probability bounded throughout. This classification turned out to be consistent with the subsequent market evolution: a recovery within two quarters, no cascade of liquidity stress, no policy-grade intervention, no credit-spread blow-out of the kind that characterises a true Crisis-regime episode.

What this case illustrates is something the binary literature on regime detection often misses: the question of “did the system detect the event” is less interesting than the question of “did the system correctly characterise the event.” Our framework discriminates between systemic panic and severe-but-bounded stress on the basis of how the return distribution deforms across multiple temporal scales simultaneously. The seven-regime taxonomy of Section 3.5 exists precisely to make this distinction available; the Hormuz episode demonstrates that the distinction is empirically meaningful and not merely a notational convenience.

5.5 Out-of-sample pipeline performance

We close the section with the numerical summary that quantitative-finance reviewers will look for first. On out-of-sample data covering January 2015 through April 2026 (2,847 trading days), the pipeline produces the following results.

For each of the three crises in the out-of-sample period—COVID 2020, Inflation 2022, and Hormuz 2025—the aggregated stress posterior reached its maximum within the 60 trading days preceding the market trough: 0.998 for COVID, 0.999 for the inflation bear market of 2022, and 1.000 for the Hormuz episode. Across the full test set, using a stress-posterior threshold of 0.30 and a ± 90 trading-day window around each crisis trough as the positive class, the system achieves a global true-positive rate of 77.4% against a false-positive rate of 35.9%. During the 2017 low-volatility year, used as a calm-period stress test, the false-positive rate is 0.0%.

A natural methodological question is whether the random-forest classifier should be coupled more tightly to the particle filter—for instance, through a convex combination $L(r) = \lambda \cdot L_{kNN}(r) + (1 - \lambda) \cdot L_{RF}(r)$ in the spirit of Grisetti et al. [2007]. We tested this. Across $\lambda \times T$ parameter sweeps spanning the natural range of mixture weights and temperatures, the geometric k -NN likelihood proved Pareto-dominant. We attribute this to a structural property: the geometric likelihood expresses distance to the empirical core of each regime in the standardised feature space, whereas the random-forest posterior expresses confidence in regime identity. Mixing the two confuses the inferential role of the filter, which is to integrate the geometry over time rather than to repeat the classifier’s instantaneous judgement. We retain the random forest and Viterbi decoding as upstream label generators, and the k -NN-based likelihood as the canonical observation model. The mixing code remains in the companion repository for reproducibility of this ablation.

Ten-seed hardening of the pipeline confirms bitwise reproducibility of these numbers; the seeds and the snapshot are committed to the repository.

6 Conclusions

This paper has proposed a particle filter framework for real-time regime inference in financial markets, grounded in a methodological transfer from probabilistic robotics and computer vision. The central contribution is the integration of particle filtering with multi-scale observation modelling: a multi-scale representation of the distributional shape of return distributions, computed across a

hierarchy of temporal windows and transferred to the particle filter as a structured prior through strategic injection.

The empirical validation rests on four major crisis episodes (Dotcom 2002, Lehman 2009, COVID 2020, Inflation 2022) and one differentiating counter-case (Hormuz 2025). Across the four crashes, the share of leading sector-eigenvalue energy (λ_1 ratio) is the only feature in our analysis with cross-crisis robustness defined as $|d| > 1$ on every episode; classical within-asset shape descriptors (realised volatility, Bowley downside skewness, tail quantile) achieve substantially larger effect sizes on most crises but vary more across episodes. On the out-of-sample period 2015–2026, the pipeline produces a global true-positive rate of 77.4% against a false-positive rate of 35.9%, with zero false positives during the 2017 low-volatility year. For the abrupt COVID-19 shock, the particle filter confirmed the regime transition twenty-one trading days before the market trough—a confirmation-detector outcome we present as a feature of the system rather than as evidence of pre-peak early warning.

The Hormuz episode of 2025 provides a different kind of validation. The framework classified this period as Bear rather than Crisis, a distinction reflecting a genuine structural difference in distributional shape. The immunisation principle does not equate all severe drawdowns; it discriminates between systemic panic and severe-but-bounded stress on the basis of how the return distribution deforms across all temporal scales simultaneously.

The results support the central claim: the richness of the observation model and the structural informativeness of particle injection are the primary enablers of early regime detection through this class of methods. We therefore propose to direct research efforts towards the construction and elaboration of a new class of indicators and oscillators specifically designed for the early detection of regime changes.

6.1 Limitations

The validation is conducted on a single index. The transition from clusters to seven labelled regime categories involves a mapping step that combines empirical cluster geometry with established market terminology; the robustness of this mapping to alternative cluster-to-regime assignments merits dedicated investigation in the future. We focus on early detection of crash onset; early detection of recovery and end-phase signals needs further work. The system does not claim pre-peak detection of fully exogenous shocks, as discussed in Section 5.3; the COVID-19 case demonstrates confirmation with substantial lead-time before the trough, not pre-peak detection. The AUC-based scoring popular in regime-detection literature is not the primary metric in our evaluation, which favours pre-peak posterior magnitude and threshold-based binary classification with explicit calm-period stress testing.

6.2 Future Research Directions

The contribution of this paper is intentionally framed as opening rather than closing. We claim a central observation—that financial regime transitions announce themselves through distributional deformation across multiple temporal scales—and demonstrate that this observation can be operationalised through particle-based inference with structurally informed observation models. We do not claim a finished theory. The instruments offered here are first instruments; the four directions sketched below are not so much ambitions as invitations.

First, cross-asset generalisation: can regime detection transfer from broad indices to individual assets? Which feature-relevance patterns differ? Do single stocks require additional distributional descriptors beyond the present feature set?

Second, sub-regime discrimination: can the framework detect corrections and distinguish them from pullbacks in real-time? This is fundamentally a binary classification task requiring finer distributional shape analysis at the boundary between Sideways and Stress regimes.

Third, agricultural futures and commodity regime detection: how do seasonal patterns interact with the multi-scale approach? Do agricultural cycles require specialised temporal windows?

Fourth, and most fundamentally: the development of a new generation of market indicators structurally sensitive to regime transitions, rather than retrospective confirmations of them. How precisely can the onset of a structural break be characterised, down to the resolution of individual reversals?

Beyond refining existing models, a productive direction of research may be to ask which aspects of market structure remain invisible to current representations—and to investigate whether building new distributional instruments might lead to better understanding of market regime transitions.

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